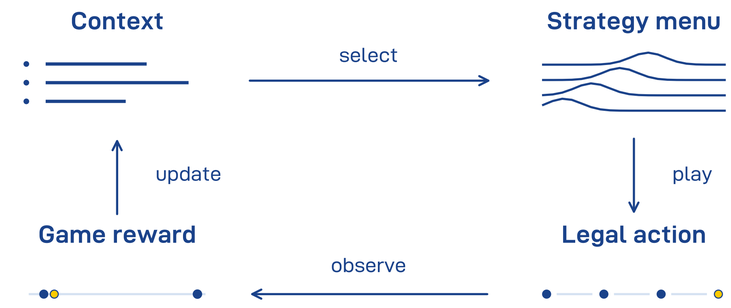
RL-Menu Agent Learns to Fold by Seat in 27,000 GLEE Games

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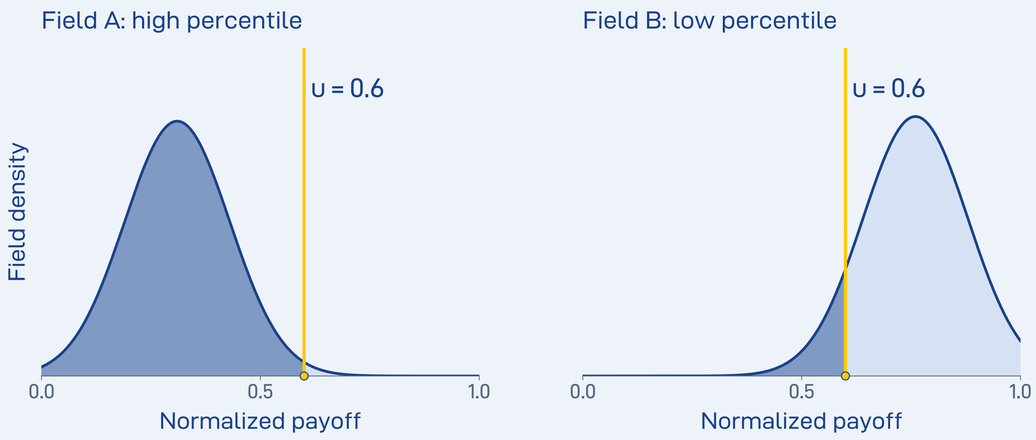


Bargaining · Negotiation · Persuasion

A menu over explicit strategies; no LLM call during play.

Learning a policy from field-relative rewards

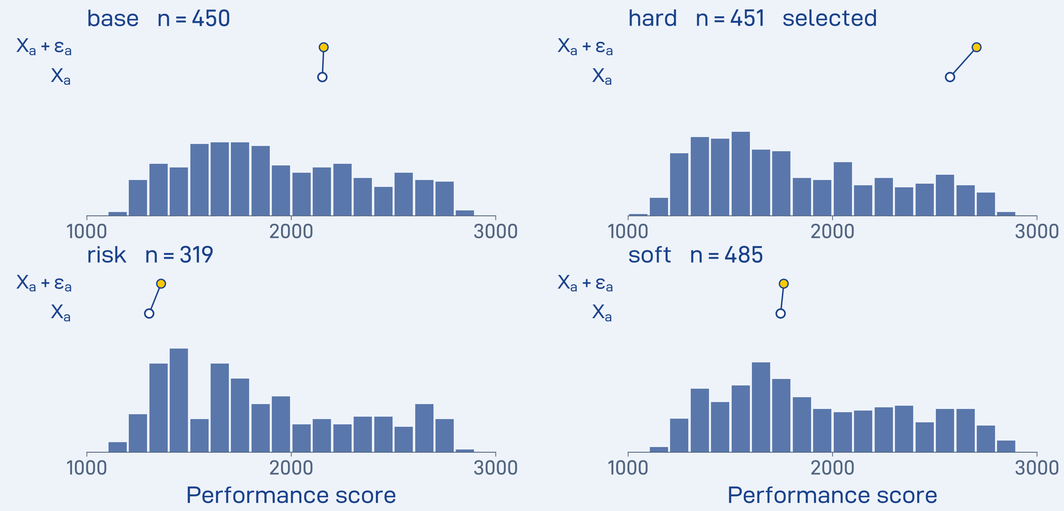
Same payoff, different field-relative position



$$\text{percentile}(u \mid c, \text{role})$$

Schematic distributions. Role and configuration stay fixed.

Observed returns, sampling and selection



$$X_a \sim \text{Unif}(D_{c,a}), \quad \varepsilon_a \sim N(0, 80^2)$$

$$a^\star = \arg\max_a (X_a + \varepsilon_a)$$

Seat-dependent strategy

We close

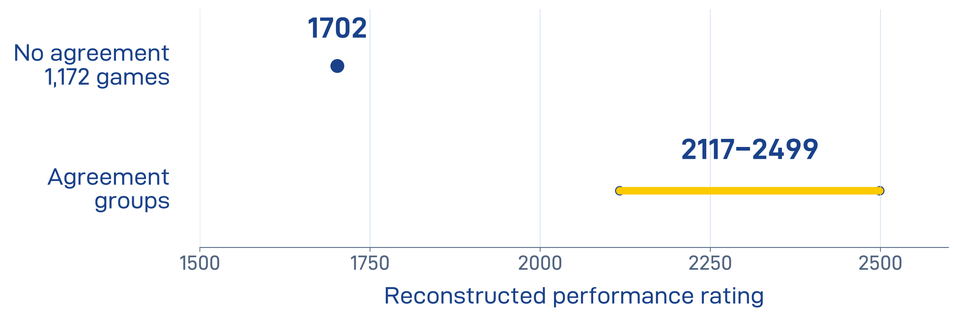


Opponent closes



Four-offer finite horizon (illustrative).

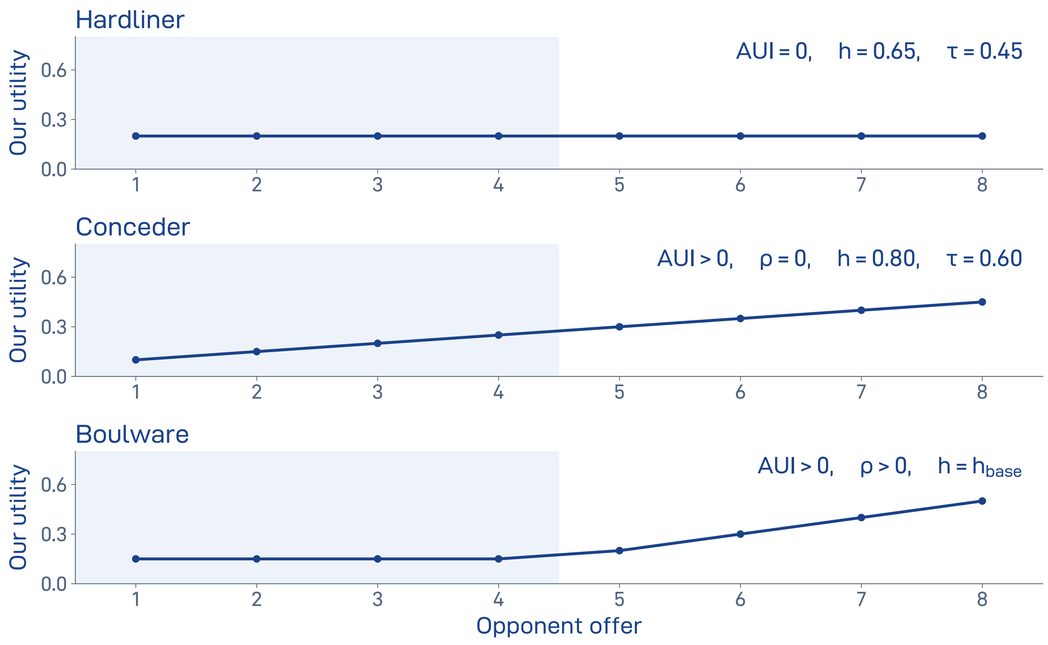
Observed when the opponent held the final offer



Outcome-group means; the range is not a confidence interval or a randomized treatment effect.

Opponent modeling

Early repetition and utility improvement



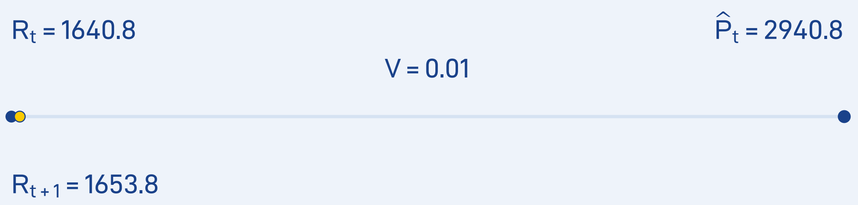
$$\rho = 1 - \frac{|\text{unique}(u_{1:m})|}{m}, \qquad m = \left\lfloor \frac{n}{2} \right\rfloor$$

Proposed surplus share h; acceptance threshold τ. Early half shaded.

Illustrative traces. AUI counts recursive mean-utility improvements. Other rules can take precedence.

Measuring behavior in a changing field

Performance from family-rating increments



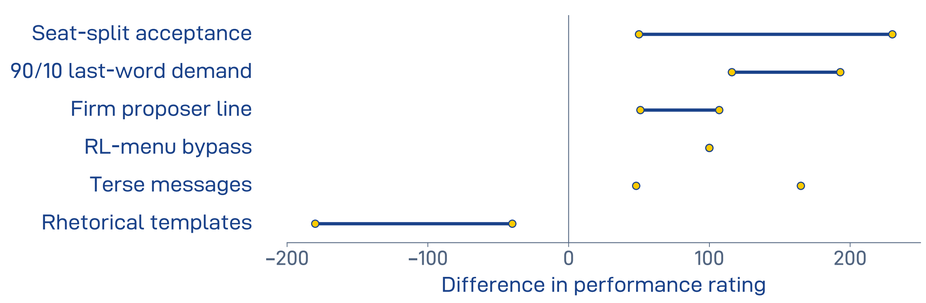
$$R_{t+1} = R_t + V(P_t - R_t), \qquad \hat{P}_t = R_t + \frac{R_{t+1} - R_t}{V}$$

$$\hat{V} = \frac{13.0}{2940 - 1640.8} \simeq 0.01$$

Single-game increment and unique same-family match. Calibration uses rounded interface values.

Reported differences across live comparisons

Final: rank 14, rating 2219.8. Peak: rank 5, rating 2302.8.



Ranges span reported comparisons, not confidence intervals.

Historical report: about 27,000 games.

Retained snapshot: 25,257 attributed games.

Concurrent live comparisons were adaptive; field composition and policy overlays changed between windows.

Shapira et al. (2024): arXiv:2410.05254
Shymanski (2025): arXiv:2512.06595
Faratin et al. (1998): RAS 24, 159–182